COMPOSITION OF FM FUNCTORS CORRESPONDS TO THE CONVOLVED KERNEL $\MATHRM{PR}_{13,*}(\MATHRM{PR}_{12}^*K \OTIMES \MATHRM{PR}_{23}^*L)$

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0FZH: sources/stacks-chow/tags.txt, source lines 19952–19952. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZD: sources/stacks-chow/tags.txt, source lines 19948–19948. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02KG: sources/stacks-chow/tags.txt, source lines 3371–3371. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0FZE: sources/stacks-chow/tags.txt, source lines 19949–19949. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let R be a ring. Let X, Y, Z be schemes over R. Assume X and Y are quasi-compact and have affine diagonal. Let*

$$F : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Y) \quad \text{and} \quad G : QCoh(\mathcal{O}_Y) \to QCoh(\mathcal{O}_Z)$$

be R-linear exact functors which commute with arbitrary direct sums. Let $\mathcal{K}$ in $QCoh(\mathcal{O}_{X \times_R Y})$ and $\mathcal{L}$ in $QCoh(\mathcal{O}_{Y \times_R Z})$ be the corresponding "kernels", see Lemma 0FZH. Then $G \circ F$ corresponds to $pr_{13,*}(pr_{12}^*\mathcal{K} \otimes_{\mathcal{O}_{X \times_R Y \times_R Z}} pr_{23}^*\mathcal{L})$ in $QCoh(\mathcal{O}_{X \times_R Z})$.

Proof. Since $G \circ F : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Z)$ is R-linear, exact, and commutes with arbitrary direct sums, we find by Lemma 0FZH that there exists an $\mathcal{M}$ in $QCoh(\mathcal{O}_{X \times_R Z})$ corresponding to $G \circ F$. On the other hand, denote $\mathcal{E} = \mathrm{pr}_{13,*}(\mathrm{pr}_{12}^*\mathcal{K} \otimes \mathrm{pr}_{23}^*\mathcal{L})$. Here and in the rest of the proof we omit the subscript from the tensor

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products. Let $U \subset X$ and $W \subset Z$ be affine open subschemes. To prove the lemma, we will construct an isomorphism

$$\Gamma(U \times_R W, \mathcal{E}) \cong \Gamma(U \times_R W, \mathcal{M})$$

compatible with restriction mappings for varying U and W.

First, we observe that

$$\Gamma(U \times_R W, \mathcal{E}) = \Gamma(U \times_R Y \times_R W, \mathrm{pr}_{12}^*\mathcal{K} \otimes \mathrm{pr}_{23}^*\mathcal{L})$$

by construction. Thus we have to show that the same thing is true for $\mathcal{M}$.

Write $U = \mathrm{Spec}(A)$ and denote $j : U \to X$ the inclusion morphism. Recall from the construction of $\mathcal{M}$ in the proof of Lemma 0FZD that

$$\Gamma(U \times_R W, \mathcal{M}) = \Gamma(W, G(F(j_*\mathcal{O}_U)))$$

where the A-module action on the right hand side is given by the action of A on $\mathcal{O}_U$. The correspondence between F and $\mathcal{K}$ tells us that $F(j_*\mathcal{O}_U) = b_*(a^*j_*\mathcal{O}_U \otimes \mathcal{K})$ where $a : X \times_R Y \to X$ and $b : X \times_R Y \to Y$ are the projection morphisms. Since j is an affine morphism, we have $a^*j_*\mathcal{O}_U = (j \times \mathrm{id}_Y)_*\mathcal{O}_{U \times_R Y}$ by Cohomology of Schemes, Lemma 02KG. Next, we have $(j \times \mathrm{id}_Y)_*\mathcal{O}_{U \times_R Y} \otimes \mathcal{K} = (j \times \mathrm{id}_Y)_*\mathcal{K}|_{U \times_R Y}$ by Remark 0FZE for example. Putting what we have found together we find

$$F(j_*\mathcal{O}_U) = (U \times_R Y \to Y)_*\mathcal{K}|_{U \times_R Y}$$

with obvious A-action. (This formula is implicit in the proof of Lemma 0FZD.) Applying the functor G we obtain

$$G(F(j_*\mathcal{O}_U)) = t_*(s^*((U \times_R Y \to Y)_*\mathcal{K}|_{U \times_R Y}) \otimes \mathcal{L})$$

where $s : Y \times_R Z \to Y$ and $t : Y \times_R Z \to Z$ are the projection morphisms. Again using affine base change (Cohomology of Schemes, Lemma 02KG) but this time for the square

$$\begin{array}{ccc} U \times_R Y \times_R Z & \longrightarrow & U \times_R Y \\ \downarrow & & \downarrow \\ Y \times_R Z & \longrightarrow & Y \end{array}$$

we obtain

$$s^*((U \times_R Y \to Y)_*\mathcal{K}|_{U \times_R Y}) = (U \times_R Y \times_R Z \to Y \times_R Z)_*\mathrm{pr}_{12}^*\mathcal{K}|_{U \times_R Y \times_R Z}$$

Using Remark 0FZE again we find

$$\begin{aligned} &(U \times_R Y \times_R Z \to Y \times_R Z)_*\mathrm{pr}_{12}^*\mathcal{K}|_{U \times_R Y \times_R Z} \otimes \mathcal{L} \\ &= (U \times_R Y \times_R Z \to Y \times_R Z)_* \left(\mathrm{pr}_{12}^*\mathcal{K} \otimes \mathrm{pr}_{23}^*\mathcal{L}\right)|_{U \times_R Y \times_R Z} \end{aligned}$$

Applying the functor $\Gamma(W, t_*(-)) = \Gamma(Y \times_R W, -)$ to this we obtain

$$\begin{aligned} \Gamma(U \times_R W, \mathcal{M}) &= \Gamma(W, G(F(j_*\mathcal{O}_U))) \\ &= \Gamma(Y \times_R W, (U \times_R Y \times_R Z \to Y \times_R Z)_*(\mathrm{pr}_{12}^*\mathcal{K} \otimes \mathrm{pr}_{23}^*\mathcal{L})|_{U \times_R Y \times_R Z}) \\ &= \Gamma(U \times_R Y \times_R W, \mathrm{pr}_{12}^*\mathcal{K} \otimes \mathrm{pr}_{23}^*\mathcal{L}) \end{aligned}$$

as desired. We omit the verification that these isomorphisms are compatible with restriction mappings. □